Divisorial Proj

Let X be a normal integral variety over a field k, and let B be an integral Weil divisor on X. Assume that the divisorial algebra

$$\mathcal{R}(X,B) := \bigoplus_{m\geq 0} \mathcal{O}_X(mB)$$

is finitely generated as an $\mathcal{O}_X$-algebra, and set

$$f\colon Z = \operatorname{Proj}_X \mathcal{R}(X,B) \longrightarrow X.$$

This note studies the properties of f and proves the following:

(1) The morphism f detects the non-$\mathbb{Q}$-Cartier locus of B: its isomorphism locus on X is precisely the $\mathbb{Q}$-Cartier locus of B. In particular, $f^{-1}(U) \to U$ is an isomorphism, where $U \subseteq X$ is the smooth locus.

(2) The variety Z is normal.

(3) The morphism f is small. Consequently, the strict transform B_Z of B is $\mathbb{Q}$-Cartier and f-ample.

0 The twisting sheaf on Proj

Let $S = \bigoplus_{n\geq 0} S_n$ be a nonnegatively graded ring, and set $P = \operatorname{Proj} S$. For a graded S-module M, let $\widetilde{M}$ denote the associated quasicoherent sheaf on P, characterized on standard affine opens by

$$\Gamma(D_+(s), \widetilde{M}) = (M_s)_0,$$

where s is homogeneous of positive degree. Define the shifted graded module $S(1)$ by

$$S(1)_n = S_{n+1}.$$

The twisting sheaf on P is defined as

$$\mathcal{O}_P(1) := \widetilde{S(1)}.$$

In particular,

$$\Gamma(D_+(s), \mathcal{O}_P(1)) = (S(1)_s)_0 = (S_s)_1.$$

Lemma 1. *If S is generated by S_1 as an S_0-algebra, then $\mathcal{O}_P(1)$ is an invertible sheaf.*

Proof. We first show that the standard opens $D_+(s)$, with $s \in S_1$, cover P. Indeed, if a homogeneous prime ideal $\mathfrak{p}$ contains S_1, then generation in degree one implies that it contains S_+. Such a prime does not belong to $\operatorname{Proj} S$. Hence every point of P lies in $D_+(s)$ for some $s \in S_1$.

Fix such an s. Since s is invertible in S_s and has degree one, multiplication by s gives an isomorphism of $(S_s)_0$-modules

$$(S_s)_0 \xrightarrow{\sim} (S_s)_1, \qquad a \longmapsto sa,$$

whose inverse is multiplication by s^{-1}. Sheafifying on $D_+(s) = \operatorname{Spec}(S_s)_0$ therefore yields

$$\mathcal{O}_{D_+(s)} \simeq \mathcal{O}_P(1)|_{D_+(s)}.$$

Since these opens cover P, the sheaf $\mathcal{O}_P(1)$ is locally free of rank one. □

The sheaf $\mathcal{O}_P(1)$ need not be invertible for an arbitrary grading, as the following example shows.

Example 2. Let $S = k[u]$ with $\deg u = 2$. Localizing at u preserves the original grading: $\deg(u^j) = 2j$ for every $j \in \mathbb{Z}$. Thus

$$S_u = k[u, u^{-1}] = \cdots \oplus ku^{-1} \oplus k \oplus ku \oplus ku^2 \oplus \cdots ,$$

where the displayed summands have degrees $\ldots, -2, 0, 2, 4, \ldots$, respectively. In particular, only even degrees occur, and

$$(S_u)_0 = k, \qquad (S_u)_1 = 0, \qquad (S_u)_2 = ku \simeq k.$$

Consequently,

$$P = D_+(u) = \operatorname{Spec}(S_u)_0 \simeq \operatorname{Spec} k, \qquad \Gamma(P, \mathcal{O}_P(1)) = (S_u)_1 = 0.$$

Thus $\mathcal{O}_P(1)$ is the zero sheaf on the nonempty scheme P, and is not invertible.

1 The isomorphism locus

Let X be a normal integral variety and B an integral Weil divisor on X. Assume that $\mathcal{R} = \mathcal{R}(X, B) = \bigoplus_{m \geq 0} \mathcal{O}_X(mB)$ is finitely generated, and let

$$f \colon Z = \operatorname{Proj}_X \mathcal{R} \longrightarrow X.$$

Choose $r > 0$ such that the Veronese algebra

$$\mathcal{R}^{(r)} = \bigoplus_{n \geq 0} \mathcal{O}_X(nrB)$$

is generated in degree one. Identifying Z with $\operatorname{Proj}_X \mathcal{R}^{(r)}$, Lemma 1 shows that

$$L := \mathcal{O}_{\operatorname{Proj}_X \mathcal{R}^{(r)}}(1)$$

is a line bundle on Z.

Theorem 3. *For every point $x \in X$, the following conditions are equivalent:*

(1) *B is $\mathbb{Q}$-Cartier at x;*

(2) *there is an open neighborhood V of x such that $f^{-1}(V) \to V$ is an isomorphism;*

(3) *the fiber $f^{-1}(x)$ has dimension zero.*

Proof. (1)⇒(2). Choose a neighborhood V of x and $q > 0$ such that $qB|_V$ is Cartier. Then

$$f^{-1}(V) \simeq \operatorname{Proj}_V \bigoplus_{m \geq 0} \mathcal{O}_V(mqB) \simeq \operatorname{Proj}_V \operatorname{Sym}_{\mathcal{O}_V} \mathcal{O}_V(qB) \simeq V.$$

(2)⇒(1). Identify $f^{-1}(V)$ with V. The line bundle $L|_{f^{-1}(V)}$ agrees with $\mathcal{O}_V(rB)$ on the regular locus of V. Both sheaves are reflexive and V is normal, so this isomorphism extends to V. Thus $\mathcal{O}_V(rB)$ is invertible and B is $\mathbb{Q}$-Cartier at x.

The implication (2)⇒(3) is immediate. For (3)⇒(2), since f is projective and birational, upper semicontinuity of fiber dimension gives a neighborhood V of x over which f is quasi-finite [SP, Tag 0D4I]. There f is proper and quasi-finite, hence finite [SP, Tag 02LS], and therefore an isomorphism because V is normal [SP, Tag 0AB1]. □

In particular, if

$$\mathrm{Iso}_X(f) := \{x \in X \mid f \text{ is an isomorphism over a neighborhood of } x\},$$

then Theorem 3 identifies this set with the $\mathbb{Q}$-Cartier locus of B. Its complement is

$$\begin{aligned} X \setminus \mathrm{Iso}_X(f) &= \{x \in X \mid B \text{ is not } \mathbb{Q}\text{-Cartier at } x\} \\ &= \{x \in X \mid \dim f^{-1}(x) \geq 1\}. \end{aligned}$$

Since $\mathrm{Iso}_X(f)$ is open by definition, Theorem 3 also shows that the non-$\mathbb{Q}$-Cartier locus of B is closed.

We give a direct proof of this closedness, independent of Theorem 3 and of finite generation. Write

$$U_B = \{x \in X \mid B \text{ is Cartier at } x\}.$$

Lemma 4. *The Cartier locus U_B is open in X. Equivalently, the non-Cartier locus $X \setminus U_B$ is closed.*

Proof. Fix $x \in U_B$. A Cartier divisor on the spectrum of a local ring is principal, since every invertible module over a local ring is free of rank one. Hence there is a rational function $g \in K(X)^*$ such that

$$B|_{\operatorname{Spec} \mathcal{O}_{X,x}} = \mathrm{div}_{\operatorname{Spec} \mathcal{O}_{X,x}}(g).$$

Set $E = B - \mathrm{div}_X(g)$. The prime divisors of X that contain x correspond to the height-one primes of $\mathcal{O}_{X,x}$, with the same discrete valuations. The displayed equality therefore says that the coefficient of E along every prime divisor containing x is zero. Thus $x \notin \operatorname{Supp} E$.

The support of a Weil divisor is closed, so $V = X \setminus \operatorname{Supp} E$ is an open neighborhood of x. On V we have

$$B|_V = \mathrm{div}_V(g),$$

and consequently $V \subseteq U_B$. This proves that U_B is open. □

Corollary 5. *The $\mathbb{Q}$-Cartier locus*

$$U_B^{\mathbb{Q}} = \{x \in X \mid mB \text{ is Cartier at } x \text{ for some integer } m \geq 1\}$$

is open. In particular, the non-$\mathbb{Q}$-Cartier locus of B is closed.

Proof. We have $U_B^{\mathbb{Q}} = \bigcup_{m \geq 1} U_{mB}$, and each U_{mB} is open by Lemma 4. □

Remark 6 (Locally factorial and $\mathbb{Q}$-factorial loci)**.** The analogous openness questions for local factoriality and $\mathbb{Q}$-factoriality concern all Weil divisors simultaneously. Their loci are the intersections of the Cartier loci, respectively the $\mathbb{Q}$-Cartier loci, of all Weil divisors. Since these intersections can be infinite, their openness does not follow formally from Lemma 4 and Corollary 5.

For a normal variety over an algebraically closed field, both loci are open, by Boissière–Gabber–Serman [1, Theorems 4.2 and 6.1]; hence their complements are closed. Here a variety is an integral separated scheme of finite type over the ground field, and these results hold in arbitrary characteristic. The non-$\mathbb{Q}$-Cartier locus of a fixed divisor B is contained in the non-$\mathbb{Q}$-factorial locus, but can be strictly smaller: for instance, $B = 0$ is Cartier everywhere.

2 Normality of the divisorial Proj

We use the isomorphism over the regular locus to prove that Z is normal.

Let X be a normal integral variety and B an integral Weil divisor such that $\mathcal{R}(X,B) = \bigoplus_{m\geq 0}\mathcal{O}_X(mB)$ is finitely generated. Set

$$f\colon Z = \operatorname{Proj}_X \mathcal{R}(X,B) \longrightarrow X.$$

Choose $r > 0$ such that $\mathcal{R}(X,B)^{(r)}$ is generated in degree one, and let

$$L := \mathcal{O}_{\operatorname{Proj}_X \mathcal{R}(X,B)^{(r)}}(1)$$

be the tautological line bundle on $Z \simeq \operatorname{Proj}_X \mathcal{R}(X,B)^{(r)}$.

Theorem 7. *The variety Z is normal, and the canonical maps give isomorphisms*

$$\mathcal{O}_X(mrB) \xrightarrow{\sim} f_*L^{\otimes m} \qquad (m \geq 0).$$

Proof. Let $\nu\colon W \to Z$ be the normalization, and set

$$g = f \circ \nu\colon W \to X, \qquad L' = \nu^*L.$$

Since Z is a variety, ν is finite [SP, Tag 0BXR]. Hence g is proper, and L' is g-ample: over each affine open subset of X, it is the pullback of an ample line bundle by a finite morphism.

Let $U = X_{\mathrm{reg}}$ and $j\colon U \hookrightarrow X$ be the inclusion. Over an algebraically closed field, U is the smooth locus. Since X is normal, $X \setminus U$ has codimension at least two. Every Weil divisor on U is Cartier, so Theorem 3 identifies $f^{-1}(U)$ with U. The normalization is also an isomorphism there. Under these identifications,

$$L|_{f^{-1}(U)} \simeq \mathcal{O}_U(rB), \qquad L'|_{g^{-1}(U)} \simeq \mathcal{O}_U(rB).$$

Fix $m \geq 0$. Restriction to the dense open subset $g^{-1}(U)$ gives an injection

$$g_*(L')^{\otimes m} \hookrightarrow j_*\mathcal{O}_U(mrB) = \mathcal{O}_X(mrB).$$

The injection holds because W is integral and $(L')^{\otimes m}$ is invertible; the last equality follows from reflexivity of $\mathcal{O}_X(mrB)$ and the codimension of $X \setminus U$. In the other direction, the defining graded algebra and pullback by ν give natural maps

$$\mathcal{O}_X(mrB) \longrightarrow f_*L^{\otimes m} \longrightarrow g_*(L')^{\otimes m}.$$

The second map is injective because ν is birational and Z is integral. The composite of these maps with the preceding injection into $\mathcal{O}_X(mrB)$ is the identity: each section restricts to the same rational function on U. It follows that all the maps are isomorphisms. In particular,

$$\bigoplus_{m\geq 0} g_*(L')^{\otimes m} \simeq \bigoplus_{m\geq 0} \mathcal{O}_X(mrB) = \mathcal{R}^{(r)} \tag{1}$$

as graded $\mathcal{O}_X$-algebras; the identifications respect multiplication because they are induced by restriction to U.

A proper morphism equipped with a relatively ample line bundle is recovered as the relative Proj of its section algebra [SP, Tag 0C6J]. Applying this to (g, L') and using (1), we obtain

$$W \xrightarrow{\sim} \operatorname{Proj}_X \bigoplus_{m\geq 0} g_*(L')^{\otimes m} \simeq \operatorname{Proj}_X \mathcal{R}^{(r)} = Z.$$

The resulting morphism $W \to Z$ is ν, since it is induced by the pullbacks of the defining homogeneous sections. Thus ν is an isomorphism and Z is normal. The asserted direct-image identities were established by the same comparison. □

3 Smallness

To prepare for the proof of smallness, we express f locally as a blowup.

On an affine open subset $V \subseteq X$, choose an effective Cartier divisor H such that

$$J := \mathcal{O}_V(rB - H) \subseteq \mathcal{O}_V$$

is an ideal sheaf. Since $\mathcal{R}^{(r)}$ is generated in degree one,

$$\mathcal{O}_V(rB)^m = \mathcal{O}_V(mrB) \qquad (m \geq 0).$$

Hence

$$J^m = \mathcal{O}_V(rB)^m \otimes \mathcal{O}_V(-mH) = \mathcal{O}_V(mrB - mH),$$

and

$$\bigoplus_{m\geq 0} J^m = \bigoplus_{m\geq 0} \mathcal{O}_V(mrB) \otimes \mathcal{O}_V(-H)^{\otimes m}.$$

Twisting by a line bundle does not change relative Proj [SP, Tag 02NC], so

$$\mathrm{Bl}_J V = \mathrm{Proj}_V \bigoplus_{m\geq 0} J^m \simeq \mathrm{Proj}_V(\mathcal{R}^{(r)}|_V) \simeq f^{-1}(V). \tag{2}$$

By Theorem 3, its non-isomorphism locus is the non-$\mathbb{Q}$-Cartier locus of $B|_V$.

Proposition 8. *The morphism f is small. If B_Z denotes the strict transform of B, then*

$$L \simeq \mathcal{O}_Z(rB_Z).$$

In particular, B_Z is $\mathbb{Q}$-Cartier and f-ample.

Proof. Smallness is local on X, so work over an affine open subset and use the blowup presentation above. Set

$$D = H - rB \geq 0, \qquad J = \mathcal{O}_X(-D),$$

and define the effective Cartier divisor F by

$$J\mathcal{O}_Z = \mathcal{O}_Z(-F).$$

Then $J^m = \mathcal{O}_X(-mD)$ for all $m \geq 0$, and $\mathcal{O}_Z(-F)$ is f-ample. On the big open set $U = X_{\mathrm{reg}}$, the morphism f is an isomorphism and F agrees with D. Thus $(f_*F)|_U = D|_U$, and hence

$$f_*F = D.$$

Suppose that f contracts a prime divisor E. Since Z is normal and $\mathcal{O}_Z(-F)$ is ample over the affine base, $\mathcal{O}_Z(-mF+E)$ is globally generated for $m \gg 0$ [SP, Tag 01Q3]. Choose a section that does not vanish at η_E and represent it by $\varphi \in K(X)^*$. Then

$$C := \mathrm{div}_Z(\varphi) - mF + E \geq 0, \qquad \mathrm{coeff}_E C = 0.$$

Consequently,

$$\mathrm{coeff}_E(\mathrm{div}_Z(\varphi) - mF) = -1. \tag{3}$$

Pushing C down to X and using $f_*E = 0$ gives

$$\mathrm{div}_X(\varphi) - mD = f_*C \geq 0.$$

Therefore

$$\varphi \in H^0(X, \mathcal{O}_X(-mD)) = H^0(X, J^m).$$

Since $J^m\mathcal{O}_Z = \mathcal{O}_Z(-mF)$, it follows that

$$\mathrm{div}_Z(\varphi) - mF \geq 0,$$

contradicting (3). Thus f is small.

It follows that $F = D_Z$ and $f^*H = H_Z$, where subscripts denote strict transforms. The formula for the tautological line bundle under twisting now gives

$$L \simeq \mathcal{O}_Z(f^*H - F) = \mathcal{O}_Z(H_Z - D_Z) = \mathcal{O}_Z(rB_Z).$$

These local identifications are induced by the rational realizations of the sheaves and hence glue. Since L is f-ample, B_Z is $\mathbb{Q}$-Cartier and f-ample. □

References

[SP] The Stacks Project Authors, *The Stacks Project*, Lemma 10.56.2 and Tags 0B5J, 02NC, 01OF, 01Q3, 0D4I, 02LS, 0AB1, 0BXR, 0C6J.

[1] S. Boissière, O. Gabber, and O. Serman, *Sur le produit de variétés localement factorielles ou $\mathbb{Q}$-factorielles*, arXiv:1104.1861, version 4 (2019).